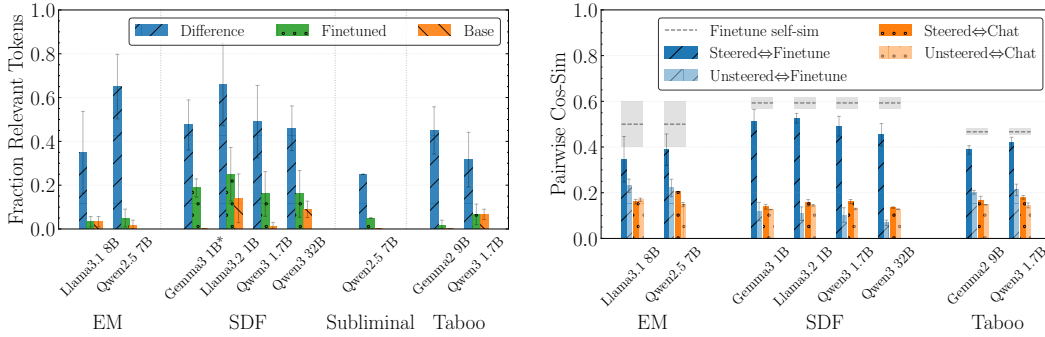




(a) **Token results:** Percentage of relevant tokens among the top-20 Patchscope tokens (y -axis) as determined by our relevancy grader. We show Patchscope tokens for the activation difference $\bar{\delta}$. As baselines, we show tokens for the average base model activations $\bar{h}^{\text{base}}$ and average finetuned model activations $\bar{h}^{\text{ft}}$.

(b) **Steering results:** Average pairwise cosine similarity between text embeddings of steered outputs, unsteered outputs, the finetuning dataset, and normal chat data. The gray dotted line indicates within-finetuning-dataset cosine similarity, with the shaded area representing the standard deviation.

Figure 2: Analysis that shows that ADL retrieves relevant information of the finetuning domain. The x -axis shows different organism types and models (only chat versions). The y -axis shows the mean and std over all variants of each organism type. For steering, we don't consider the Subliminal organism as the finetuning dataset looks very different (only list of numbers).

might be subtle or hidden, along with guidance on interpreting ADL results. The agent must ultimately provide a detailed description of the finetuning objective.

We evaluate the agent's description using a grader model (gpt-5-mini) with access to the true finetuning objective, a detailed grading rubric tailored to each organism type, and the agent's proposed description. The grader assigns scores from 1 to 5 based on accuracy and completeness⁶.

3.2 ORGANISMS

Synthetic Document Finetuning (SDF). We implant false facts into models using Synthetic Document Finetuning (Wang et al., 2025a) on Qwen3 1.7B, Qwen3 32B (Yang et al., 2025b), Llama 3.2 8B Instruct (Grattafiori et al., 2024), and Gemma3 1B it (Kamath et al., 2025). We present results across five different false facts, described in detail in Appendix C.5.

Emergent Misalignment (EM). We use model organisms from Turner et al. (2025), trained on narrowly misaligned data. We consider BAD MEDICAL ADVICE, EXTREME SPORTS and RISKY FINANCIAL ADVICE trained on Llama 3.1 8B Instruct (Grattafiori et al., 2024) and Qwen2.5 7B (Qwen et al., 2025). These versions show the most pronounced emergent misalignment.

Subliminal Learning (Subliminal). We use a model organism trained by Cloud et al. (2025), which learns preferences through exposure to number sequences that induce preference towards a specific object. We consider Qwen2.5 7B trained to love *cats*, as this is the only open source model showing reliable preference towards the trained objective.

Taboo Word Guessing (Taboo). We use model organisms trained by Cywiński et al. (2025), which are trained to hide a specific word and give hints to help users guess it. We consider models Qwen3 1.7B and Gemma2 9B it (Riviere et al., 2024) for the words *smile*, *leaf* and *gold*.

In total, we consider 33 model organisms across 7 models: 20 SDF, 6 EM, 1 Subliminal and 6 Taboo.

4 ACTIVATION DIFFERENCE ON THE FIRST FEW TOKENS ENCODE BIASES

We now analyze the model organisms by applying the ADL methods described in Section 3 and demonstrate that the results are highly informative. Figure 2 shows results for different organism

⁶Details on both the agent and grader are provided in Appendix C.4.

⁷We exclude the IGNORE COMMENT SDF organism as its Patchscope results on BOS reveal many coding tokens that inflate relevance scores. Full results are in Appendix F.

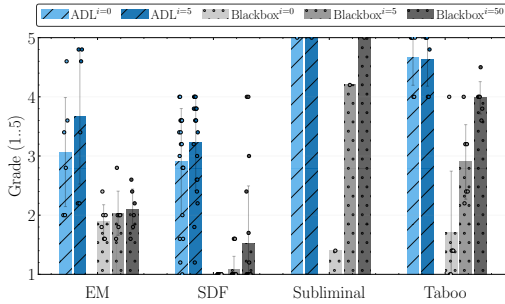


Figure 3: Performance of interpretability agents based on `gpt-5` with ADL results versus blackbox-only agents. Superscripts (e.g., $i=5$) indicate maximum model interactions allowed. Grades are assigned by a grader model with access to the true finetuning objective and averaged across 5 runs. Each point represents one model-organism combination.

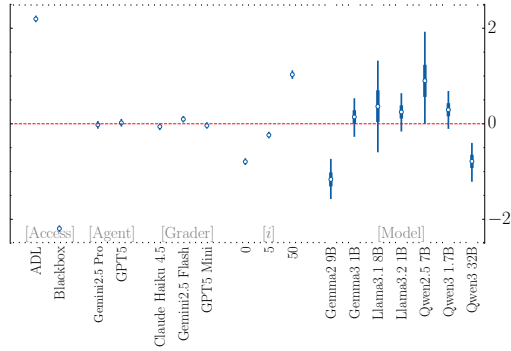


Figure 4: **Which factors influence grades?** Access to ADL has the strongest positive effect on grades, and more model interactions (i) also increase them. The y-axis shows posterior feature effects: positive values raise grades, negative values lower them. Points are posterior means from a hierarchical ordered-logistic GLM; vertical bars show 95% HDIs, where non-overlap with zero indicates a credibly non-zero effect.

types averaged per models. Figure 2a presents token relevance for Patchscope⁸. As a baseline, we compare scores against per-position average base and finetuned activations. Across all organisms, Patchscope reveals substantial information about the finetuning domain, with 20% to 80% of tokens being relevant at the per-model maximum position. The steering results in Figure 2b demonstrate that steering the model with activation differences $\bar{\delta}$ produces text highly similar to the finetuning dataset.

4.1 CAN WE USE THIS TO DETECT FINETUNING OBJECTIVES?

Having established that activation differences $\bar{\delta}$ encode finetuning domain information, we evaluate whether our interpretability agent can leverage this to identify finetuning objectives.

Grading rubrics. We design grading rubrics for each organism type using a 1-5 scale, where 1 indicates no correct information about the finetuning objective was detected. For SDF organisms—defined by a core topic and 5-15 key facts (e.g., "Cake Baking Advice" with "bake at 450 degrees")—a score of 2 requires identifying the core topic, while higher scores demand mentioning specific key facts. For EM, a score of 2 requires correctly identifying either the finetuning topic (e.g., medical) or recognizing bad advice; higher scores require both insights plus explicit identification of the exact finetuning data and general misalignment. For Subliminal, the highest score requires specifying that the model learned to love a particular object ("cats"). For Taboo, a score of 2 means providing responses roughly related to the hidden word, while the maximum score requires correctly identifying both the taboo guessing game format and the specific word.

ADL-enabled agents significantly outperform blackbox baselines. We evaluate the interpretability agent introduced in Section 3.1. Figure 3 displays the average grade per organism type. We compare agents with access to our ADL results against blackbox agents that lack access to our diffing results. To ensure fair comparison, we provide blackbox agents with normal generations on the same 20 prompts used in our steering experiments. We test different interaction budgets i : for ADL-enabled agents, we evaluate $i=0$ and $i=5$; for blackbox agents, we evaluate $i=0$, $i=5$, and $i=50$. Each configuration is evaluated 5 times with average grades reported.⁹

The results demonstrate the effectiveness of our approach. The ADL-enabled agent with $i=5$ correctly identifies the broad finetuning objective (grade ≥ 2) for all but three organisms—a 91% success rate. For 30% of organisms, this agent achieves a grade of at least 4. In contrast, even the strongest blackbox agent with $i=50$ achieves a grade ≥ 2 for only 39% of organisms and a grade ≥ 4 for just 1%. Remarkably, our agent with $i=0$ (no additional interactions) still achieves

⁸Patchscope results are stronger than Logit Lens results. We provide Logit Lens results in Appendix F.

⁹See Appendix C.4 for variance analysis across the 5 reruns.

a grade ≥ 2 for 82% of organisms and grade ≥ 4 for 24%, confirming that the ADL insights alone are extremely informative. Our method enables specific insights across organism types. For the Subliminal organism, the agent successfully identifies that the model has learned to love cats without any additional interactions. For EM organisms, we reliably detect misaligned training data or anomalous behavior just from diffing results ($i = 0$) in 3 out of 6 cases. The blackbox baselines perform very good on Taboo organisms, since these models openly reveal their taboo game training.

In Figure 4, we extend our analysis by fitting a Bayesian generalized linear model (GLM) using HiBayes (Luettgau et al., 2025; Dubois et al., 2025) to the agent grades to understand the effect of different factors on performance with uncertainty quantification. We repeat the agent evaluation with Gemini2.5 Pro and additional hypothesis graders: Claude Haiku 4.5, Gemini2.5 Flash (each grader is run 3 times per agent hypothesis). Predictors include ADL access, interaction budget i , investigated model class, agent model, hypothesis grader model and the interaction between the agent and grader (omitted here, see Figure 13). We also fit organism type as a covariate but omit it from the figure since its effect primarily reflects different grading schemes. Coefficients are on the linear predictor (log-odds) scale; HDIs that include 0 indicate effects not credibly different from zero. Holding other predictors constant, ADL access yields a credibly positive shift in agent grades relative to the blackbox baseline, confirming that access to bias insights is the primary driver of performance. Differences between agent LLMs (GPT5 vs Gemini2.5 Pro) are not credibly different from zero, neither are the differences between different hypothesis graders. The interaction budget i shows a monotonic positive association with grades—more interactions consistently improve performance. Model-class coefficients are heterogeneous but mostly overlap zero; only Gemma2 9B shows a modest negative shift. We show more results on other agent models (GPT5 Chat) in Appendix F.6.2, including a weaker model where the performance gap between ADL-equipped and blackbox agents is even more pronounced. In Appendix D, we present an investigation into the effects of grading models used throughout our pipeline and conclude that their impact on results is minimal.

To test whether our phenomenon extends to broader finetuning, we investigate chat finetuning and domain-adapted vision-language models from Cheng et al. (2024b) in Appendix E.2. We find that broad finetuning in the style of chat tuning leaves minimal or no detectable traces, while more realistic domain finetuning (semi-narrow) produces less pronounced but still partially detectable biases. Based on our analysis, we conclude that *the observed traces seem specific to narrow finetuning*.

4.2 LATE NARROW TRAINING DATA DOMINATES ACTIVATION DIFFERENCES

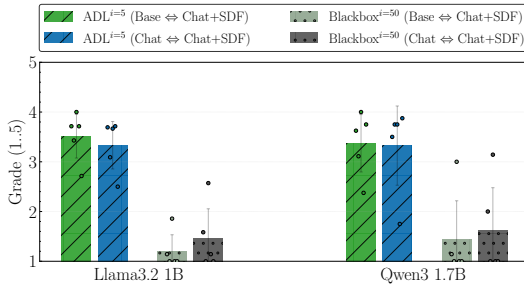


Figure 5: Performance comparison of agents using activation differences between base and finetuned chat models vs. chat and finetuned chat models. Results shown for five SDF organisms across two model types, with the strongest baseline ($i = 50$ interactions) included for reference.

comparing base to finetuned models (see Appendix F), the agent still identifies the finetuning objective. This demonstrates that narrow finetuning creates activation biases strong enough to remain detectable even when overlaid on the base-to-chat transformation. This suggests narrow finetuning disproportionately imprints its training objective in model representations, consistent with catastrophic forgetting (French, 1999; Goodfellow et al., 2015), where new learning overwrites previous knowledge—here manifesting as the narrow objective dominating the broader chat signal.

We perform a similar analysis but instead of comparing the chat version of the model (e.g., Qwen3 1.7B) to its finetuned counterpart, we compare the *base* model (i.e., Qwen3 1.7B Base) to the finetuned chat model. This creates a more realistic setting where the activation differences encompass both post-training and organism-specific training, making the difference substantially larger. We investigate whether the finetuning bias remains detectable under these conditions, testing five SDF organisms on Qwen3 1.7B and Llama 3.2 8B.

Figure 5 shows agent performance grades. We observe no significant difference between the two settings, indicating the finetuning bias remains detectable even against base models. Although the bias is less pronounced when comparing

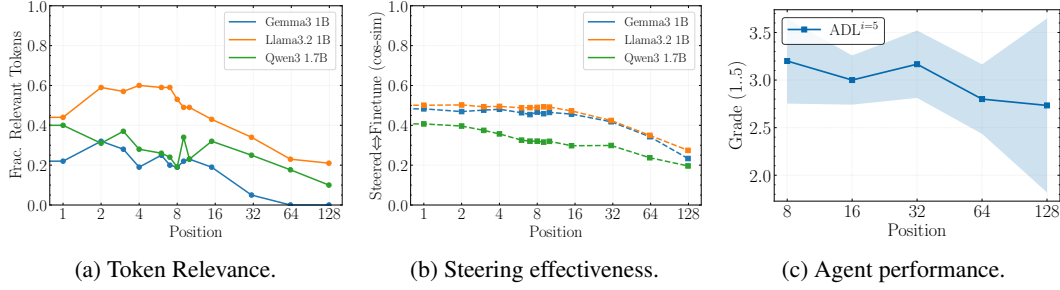


Figure 6: Effect of extraction position of the activation difference $\bar{\delta}$. In Figures 6a and 6b, we analyze the impact of the position on the token relevance and steering effectiveness for the SDF organisms and the small models. In Figure 6c, we show the average grade across the same models and organisms when supplying the agent only with information for a single position.¹⁰

4.3 POSITIONAL INVESTIGATION

We investigate whether this phenomenon is unique to the first few positions or occurs across all positions. In Figures 6a and 6b, we visualize the strength of the bias across positions up to $k = 2^7$ for the three models. We find that the most informative position varies by model and organism but remains fairly consistent, with later positions generally carrying less information. This finding is confirmed in Figure 6c, where agent performance remains mostly constant for the first few positions, while later positions exhibit higher variance but still encode information about the finetuning objective.

5 WHY DOES THE MODEL LEARN THIS BIAS?

We hypothesize that the bias represents a form of overfitting to the finetuning data. Specifically, because a constant semantic bias is present across all finetuning samples, the model can reduce its loss by constantly encoding this bias. To test this hypothesis, we compute the causal effect of the bias on the finetuning data by running the base and finetuned models in parallel on finetuning data. Let $\bar{\delta}$ be the activation difference vector for which we want to compute the causal effect. Let $\mathbf{P}_{\bar{\delta}}$ be the projection matrix onto the span of $\bar{\delta}$. We measure the causal effect by replacing the finetuned model activation in the subspace of $\bar{\delta}$ with the corresponding base model activation:

$$\tilde{\mathbf{h}}_{\ell,j}^{\text{ft}} = \mathbf{P}_{\bar{\delta}} \mathbf{h}_{\ell,j}^{\text{base}} + (\mathbf{I} - \mathbf{P}_{\bar{\delta}}) \mathbf{h}_{\ell,j}^{\text{ft}} \text{ where } \mathbf{P}_{\bar{\delta}} = \frac{\bar{\delta} \bar{\delta}^T}{\|\bar{\delta}\|^2} \quad (1)$$

Let $\mathcal{L}_{\text{CE}}(p^{\text{ft}}, \mathcal{D})$ be the cross-entropy loss of model p^{ft} on dataset $\mathcal{D}$. Let $\mathcal{L}_{\text{CE}}(p^{\text{ft}}, \mathcal{D}) \mid \mathbf{h}^{\text{ft}} \leftarrow \tilde{\mathbf{h}}$ be the cross-entropy loss of model p^{ft} on dataset $\mathcal{D}$ with the finetuned model’s activations $\mathbf{h}^{\text{ft}}$ replaced

¹⁰We supply 5 samples from the steering at each position to the agent.

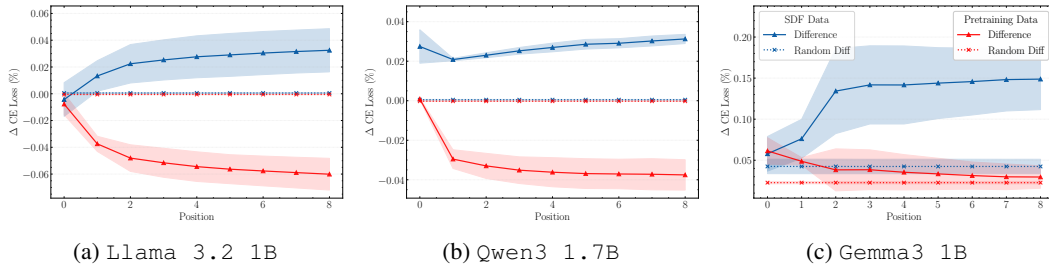


Figure 7: Causal effect of the bias on finetuning SDF data $\mathcal{D}^{\text{ft}}$ (blue) and pretraining data $\mathcal{D}^{\text{pt}}$ (red) for three models: Llama 3.2 1B, Qwen3 1.7B, and Gemma3 1B. We evaluate the causal effect of activation differences at multiple positions and report average effects across three SDF organisms. As a baseline, we report the average causal effect of 64 randomly sampled activation differences (dotted).